

Phase 1

Brainstorming & Ideation Phase

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Background

Rahul is a fitness enthusiast who exercises regularly but records his workouts manually using notebooks and spreadsheets. Although he maintains consistency in his fitness routine, he finds it difficult to organize workout history, monitor progress, and determine whether his exercise habits align with his fitness goals. He also lacks professional guidance when deciding which workouts are best suited for his current fitness level.

Problem

- Difficulty maintaining a centralized record of workouts.
- Manual tracking leads to inaccurate workout history.
- No secure user authentication for personal fitness records.
- Unable to search previous workouts efficiently.
- No personalized workout recommendations.
- No insights regarding overall workout performance.
- Time-consuming manual calculation of calories burned and workout duration.

Solution

The **FitTrack AI** provides a centralized backend system where users can securely register, authenticate, and manage workout records. Users can add, update, delete, retrieve, and search workouts through RESTful APIs. By integrating **Google Gemini AI**, the application analyzes user information and workout statistics to generate intelligent workout recommendations and personalized fitness insights.

The backend follows industry-standard security practices including JWT authentication, password hashing, middleware-based authorization, centralized error handling, and MongoDB schema validation.

User Usage

Users can register and log in with their email and password to receive a JWT token, granting access to protected APIs to:

- Create, view, update, and delete workout records, or retrieve specific workouts by ID.
- Search and filter workouts by name, category, or date.
- Generate personalized, AI-powered workout recommendations.
- Obtain AI-generated fitness insights based on workout statistics.

AI Usage

The AI module connects to Google Gemini through a dedicated service layer to deliver two core capabilities:

- **Workout Recommendations:** Processes user inputs (age, fitness goal, and experience level) to generate personalized workout plans, weekly exercise suggestions, difficulty-tailored recommendations, motivational guidance, and safety tips.
- **Fitness Insights:** Analyzes workout metrics (total workouts, average duration, and total calories burned) to produce personalized performance insights.

Outcome

The FitTrack AI backend uses a layered Model-View-Controller (MVC) architecture that separates routing, middleware, business logic, database operations, and AI services into independent modules with single responsibilities. Incoming HTTP requests to the Express server are directed to specific routes, passing through JWT authentication middleware on protected endpoints before reaching the controllers. The controllers then coordinate with Mongoose models for data persistence and a dedicated Google Gemini AI service layer to process requests and return structured JSON responses, ensuring high security, maintainability, and scalability.